



Generative AI:

How Machines Create Art, Music and Text

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Introduction

- Generative AI refers to algorithms that can generate new content.
- These algorithms are designed to produce text, images, music and art.
- Unlike traditional AI systems that primarily analyze and classify existing data.
- Generative AI synthesizes original outputs based on learned patterns from extensive datasets.

Generative AI Techniques

- 1) Generative Adversarial Networks (GANs)
 - A generator
 - A discriminator
- 2) Variational Autoencoders (VAEs)
 - The encoder
 - The decoder
- 3) Transformed-based Models

Generative Adversarial Networks (GANs)

- Developed by Ian Goodfellow and his colleagues in 2014.
- Consists of two neural networks:

1) The Generator

- Role is to create new content based on random input.
- Aim to produce outputs that are identical from real data.
- Learns to create increasingly realistic outputs.

2) The Discriminator

- Evaluates the authenticity of the data generated by the generator.
- Provide feedback to the generator, helping it to enhance its performance over time.

Challenges:

- Mode collapse: generator starts producing the same output repeatedly instead of diverse set of images.
- Require a large amount of training data to perform effectively.

Application of GANs

GANs have been used in various creative applications including:

- **Art Generation:** Artists utilize to create unique artworks that blend different style.
- **Image Creation:** Generate high resolution images from low resolution inputs.
- **Fashion Design:** GANs are employed to create novel clothing designs and patterns.

Variational Autoencoders (VAEs)

- Combines traditional autoencoders with probabilistic graphical model.
- Consist of two main components:

1) The encoder

- Takes the input data and compresses it into lower dimensional latent space.

2) The decoder

- Takes latent variable and reconstruct the data back into its original forms.

Applications of VAEs

- **Image Generation:** Allows for the creation of novel visual content.
- **Music Generation:** Analyze musical compositions and generate new pieces that capture the essence of the original data.
- **Text Generation:** Create new written content, learning the structures and styles present in existing texts.

Transformer-based Models

- Transformer is a neural network that learns the context of sequential data and thus generates meaning by tracking relationships
- Transformers operate on the principle of attention - means they can focus on different parts of the input data depending on the task at hand.

- **GPT Model:** transformer based AI model that reads and writes text like humans do, write essays, poems or emails, translate languages, answer questions
- **DALL -E:** transformer based AI model that specializes in generating images from text descriptions, create original artwork that matches your description, demonstrate stylistic variations.
- **Music LM:** transformer based AI model that specializes in generating music from text descriptions, creates music based on mood or genre, make soundtracks from descriptions



THANK YOU

